

Assignment 1 IPv4 Addressing Foundations

Networking Fundamentals

Assessment	Individual assignment	Total	50 marks
Suggested time	60 minutes	Due date	As posted in Moodle
Submission	One PDF or DOCX	Topic	Classful IPv4 addressing
Filename	Surname_Firstname_A1	Attempts	Follow Moodle settings

Purpose

Complete this assignment to show that you can recognize IPv4 address classes, apply default subnet masks, identify network and broadcast addresses, and check whether a default gateway belongs to the correct network.

Instructions

- Answer every question and show your working where a calculation is required.
- Use the classful A, B, and C rules taught in the lesson. Do not apply custom CIDR or VLSM subnetting in this assignment.
- Network and broadcast addresses cannot be assigned to devices.
- You may use a calculator, but your submitted answers must be your own work.
- Type your answers in this document or write clearly on a printed copy.
- Submit one PDF or DOCX file through the Moodle assignment link by the posted deadline.

Student Details

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Class group	TEAM-12	Submission date	27/09/2026

Mark Allocation

Section	What is assessed	Marks
Part A	Core concepts	10
Part B	Classful address identification	15
Part C	Network and broadcast analysis	15
Part D	Applied troubleshooting	10
	Total	50

Part A Core Concepts

Answer each question in one to four sentences. Each question is worth 2 marks.

- 1 Define an IPv4 address and explain why the address must be unique on the same network. [2 marks]

An IPv4 address is a number used to identify a device on a network. Every device on the same network must have a different address so that data goes to the correct device.

- 2 State one difference between flat and hierarchical addressing. Give one example of each type. [2 marks]

Flat addressing does not show where a device is located on a network. Hierarchical addressing includes network information. Examples are: flat addressing: MAC address, hierarchical addressing: IPv4 address

- 3 Compare static and dynamic IP addressing. Name the protocol normally used to assign an address dynamically. [2 marks]

A static IP address is entered manually and normally stays the same. A dynamic IP address is given automatically and can change. The protocol used is DHCP (Dynamic Host Configuration Protocol).

- 4 State one difference between classful and classless addressing. What does CIDR stand for? [2 marks]

Classful addressing uses fixed classes such as A, B, and C. Classless addressing allows flexible subnet sizes. The CIDR stands for: Classless Inter-Domain Routing.

- 5 List the four IPv4 configuration items presented in the lesson for normal name-based Internet access. [2 marks]

1) IP Address (2) Subnet Mask (3) Default Gateway (4)DNS Server

Part B Classful Address Identification

For each address, identify the class, write the network and host pattern, and give the default subnet mask. Each row is worth 3 marks.

IPv4 address	Class	Network and host pattern	Default subnet mask
23.14.6.9	A	N.H.H.H	255.0.0.0
145.80.12.200	B	N.N.H.H	255.255.0.0
198.51.100.27	C	N.N.N.H	255.255.255.0
9.200.10.1	A	N.H.H.H	255.0.0.0
188.44.9.72	B	N.N.H.H	255.255.0.0

Part C Network and Broadcast Analysis

Use the default classful mask for each address. Complete all five requested values. Each address is worth 5 marks.

1 34.72.6.19

Network address	34.0.0.0	Broadcast address	34.255.255.255
First usable host	34.0.0.1	Last usable host	34.255.255.254
Maximum usable hosts	16,777,214		
	34 is a Class A address, Default mask = 255.0.0.0, Hosts = $2^{24} - 2 = 16,777,214$		

2 150.20.7.200

Network address	150.20.0.0	Broadcast address	150.20.255.255
First usable host	150.20.0.1	Last usable host	150.20.255.254
Maximum usable hosts	65,534		
	150 is a class B address. Default mask = 255.255.0.0, Hosts = $2^{16} - 2 = 65,534$		

3 203.0.113.44

Network address	203.0.113.0	Broadcast address	203.0.113.255
First usable host	203.0.113.1	Last usable host	203.0.113.254
Maximum usable hosts	254		
	203 is a class C Address. Default mask = 255.255.255.0, Hosts = $2^8 - 2 = 254$		

Part D Applied Troubleshooting

Read the scenario, then answer all five questions.

A workstation in a training room has the following IPv4 settings. The workstation can communicate with some local devices but cannot reach resources on another network.

Setting	Configured value
IPv4 address	201.110.213.28
Subnet mask	255.255.255.0
Default gateway	201.110.214.1
DNS server	8.8.8.8

- 1 Give the network address and broadcast address for the workstation. [2 marks]

Network Address: 201.110.213.0, Broadcast Address: 201.110.213.255.

Subnet mask = 255.255.255.0, Network = 201.110.213.0, Broadcast = 201.110.213.255

- 2 Is the configured default gateway valid for this workstation? Explain your answer. [2 marks]

NO, because the gateway is 201.110.214.1, but the computer is on network 201.110.213.0. The gateway must be on the same network as the computer.

- 3 Give one valid default gateway address. Explain why the network and broadcast addresses cannot be used as the gateway. [2 marks]

A valid gateway is 201.110.213.1. The network address (201.110.213.0) cannot be used because it represents the network. The broadcast address (201.110.213.255) cannot be used because it is used to send messages to all devices on the network.

- 4 With the incorrect gateway, what communication may still work and what communication is likely to fail? [2 marks]

Devices on the same local network may still communicate. Access to other networks and the Internet will probably fail because the gateway is wrong.

5 A lab contains 25 student computers. Recommend static or dynamic addressing for the computers and name the service that would provide the addresses. [2 marks]

I recommend dynamic addressing for the 25 computers. A DHCP server can automatically give IP addresses to all computers, making management easier.